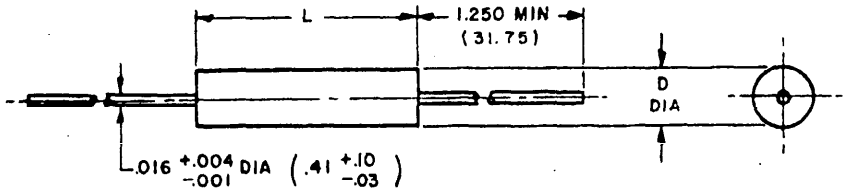
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC-DIELECTRIC (GENERAL PURPOSE) STYLES CK31, CK31.....TX, CK32, AND CK32.....TX

This specification is approved for use by Engineering and Technical Support Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.

Ⓑ THIS SPECIFICATION INACTIVE FOR DESIGN AFTER 6 JULY 1977



DIMENSIONS				
STYLE	L		D	
	INCHES	MM	INCHES	MM
CK31 CK31....TX	.250 MAX .230 MIN	6.35 MAX 5.84 MIN	.100 MAX .080 MIN	2.54 MAX 2.03 MIN
CK32 CK32... TX	.250 MAX .230 MIN	6.35 MAX 5.84 MIN	.140 MAX .120 MIN	3.56 MAX 3.05 MIN

NOTES:

1. All dimensions in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Styles CK31, CK31.....TX, CK32, and CK32.....TX Capacitors

REQUIREMENTS:

Design and construction:
Dimension and configuration - See figure 1.
Case type - Tubular with axial leads.
Capacitance value - See table I.
Capacitance tolerance - See table I.
Rated temperature - -55° to +125°C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of d-c working voltage.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).
Test potential - 150 percent rated voltage.

Ⓑ denotes change.

FSC 5910

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TABLE I. Styles CK31.....TX, CK32, and CK32.....TX

TYPE DESIGNATION 1/	DC RATED VOLTAGE	OPERATING TEMPERATURE RANGE AND VOLTAGE TEMPERATURE LIMITS	CAPACITANCE (pf)	CAPACITANCE TOLERANCE 2/
CK31BT271-	50	BT	270	J, K
CK31BT271-TX	50	BT	270	J, K
CK31BT331-	50	BT	330	J, K
CK31BT331-TX	50	BT	330	J, K
CK31BT391-	50	BT	390	J, K
CK31BT391-TX	50	BT	390	J, K
CK31BT471-	50	BT	470	J, K
CK31BT471-TX	50	BT	470	J, K
CK31BT561-	50	BT	560	J, K
CK31BT561-TX	50	BT	560	J, K
CK31BT681-	50	BT	680	J, K
CK31BT681-TX	50	BT	680	J, K
CK31BT821-	50	BT	820	J, K
CK31BT821-TX	50	BT	820	J, K
CK31BT102-	50	BT	1,000	J, K
CK31BT102-TX	50	BT	1,000	J, K
CK31BT122-	50	BT	1,200	J, K
CK31BT122-TX	50	BT	1,200	J, K
CK31BT152-	50	BT	1,500	J, K
CK31BT152-TX	50	BT	1,500	J, K
CK31BT182-	50	BT	1,800	J, K
CK31BT182-TX	50	BT	1,800	J, K
CK31BT222-	50	BT	2,200	J, K
CK31BT222-TX	50	BT	2,200	J, K
CK31BT272-	50	BT	2,700	J, K
CK31BT272-TX	50	BT	2,700	J, K
CK31BT332-	50	BT	3,300	J, K
CK31BT332-TX	50	BT	3,300	J, K
CK31BT392-	50	BT	3,900	J, K
CK31BT392-TX	50	BT	3,900	J, K
CK31BT472-	50	BT	4,700	J, K
CK31BT472-TX	50	BT	4,700	J, K
CK31BT562-	50	BT	5,600	J, K
CK31BT562-TX	50	BT	5,600	J, K
CK31BT682-	50	BT	6,800	J, K
CK31BT682-TX	50	BT	6,800	J, K
CK31BT822-	50	BT	8,200	J, K
CK31BT822-TX	50	BT	8,200	J, K
CK31BT103-	50	BT	10,000	J, K
CK31BT103-TX	50	BT	10,000	J, K
CK31BU123-	50	BU	12,000	K, M
CK31BU123-TX	50	BU	12,000	K, M
CK31BU153-	50	BU	15,000	K, M
CK31BU153-TX	50	BU	15,000	K, M
CK31BU183-	50	BU	18,000	K, M
CK31BU183-TX	50	BU	18,000	K, M
CK31BU203-	50	BU	20,000	K, M
CK31BU203-TX	50	BU	20,000	K, M

See footnote at end of table.

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TABLE I. (Cont' d)

TYPE DESIGNATION 1/	DC RATED VOLTAGE	OPERATING TEMPERATURE RANGE AND VOLTAGE TEMPERATURE LIMITS	CAPACITANCE (pf)	CAPACITANCE TOLERANCE 2/
CK31BV223-	50	BV	22,000	K, M
CK31BV223-TX	50	BV	22,000	K, M
CK31BV273-	50	BV	27,000	K, M
CK31BV273-TX	50	BV	27,000	K, M
CK31BV333-	50	BV	33,000	K, M
CK31BV333-TX	50	BV	33,000	K, M
CK31BV393-	50	BV	39,000	K, M
CK31BV393-TX	50	BV	39,000	K, M
CK31BV473-	50	BV	47,000	K, M
CK31BV473-TX	50	BV	47,000	K, M
CK31BV513-	50	BV	51,000	K, M
CK31BV513-TX	50	BV	51,000	K, M
CK32BT123-	50	BT	12,000	J, K
CK32BT123-TX	50	BT	12,000	J, K
CK32BT153-	50	BT	15,000	J, K
CK32BT153-TX	50	BT	15,000	J, K
CK32BT183-	50	BT	18,000	J, K
CK32BT183-TX	50	BT	18,000	J, K
CK32BT203-	50	BT	20,000	J, K
CK32BT203-TX	50	BT	20,000	J, K
CK32BU223-	50	BU	22,000	K, M
CK32BU223-TX	50	BU	22,000	K, M
CK32BU273-	50	BU	27,000	K, M
CK32BU273-TX	50	BU	27,000	K, M
CK32BU333-	50	BU	33,000	K, M
CK32BU333-TX	50	BU	33,000	K, M
CK32BU393-	50	BU	39,000	K, M
CK32BU393-TX	50	BU	39,000	K, M
CK32BV563-	50	BV	56,000	K, M
CK32BV563-TX	50	BV	56,000	K, M
CK32BV623-	50	BV	62,000	K, M
CK32BV623-TX	50	BV	62,000	K, M
CK32BV683-	50	BV	68,000	K, M
CK32BV683-TX	50	BV	68,000	K, M
CK32BV753-	50	BV	75,000	K, M
CK32BV753-TX	50	BV	75,000	K, M
CK32BV823-	50	BV	82,000	K, M
CK32BV823-TX	50	BV	82,000	K, M
CK32BV913-	50	BV	91,000	K, M
CK32BV913-TX	50	BV	91,000	K, M
CK32BV104-	50	BV	100,000	K, M
CK32BV104-TX	50	BV	100,000	K, M

1/ Complete type designation will include an additional symbol to indicate capacitance tolerance.

2/ Capacitance tolerance in percent: J, ±5%, K, ±10%, M, ±20%.

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Insulation resistance (IR): Method 302 of MIL-STD-202, 50 volts. 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency - 1 kHz ±100 Hz.
Test voltage - 1.0 ±0.2 volt rms.

Dissipation factor (DF): When measured at 1 kHz ±100 Hz with 1.0 ±0.2 volts rms, DF shall not be greater than the values specified in table II herein.

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50 G's).

Vibration, high frequency: Method 204 of MIL-STD 202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D.
Immersion: Method 104 of MIL-STD-202, condition B.
DWV - 250 percent of rated voltage.
IR - 10,000 megohms, min or 100 megohm-microfarads, whichever is less.
Cap. - Change not more than 2 percent for voltage temperature limits T and U, and 5 percent for voltage-temperature limit V.
DF - Not greater than initial requirement.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, capacitance change, and DF requirements same as for temperature cycling and immersion.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 . MIL-STD-202, condition B.
IR - 100,000 megohms, min or 1,000 megohm-microfarads, whichever is less.
Δ Cap. - ±5 percent of initial measurement.
DF - Not greater than initial requirement.

Voltage-temperature limits: In accordance with MIL-C-11015 except capacitance change shall be as specified in table II herein.

Life: Method 108 of MIL-STD-202.
For qualification - condition F (2,000 hours).
For quality conformance inspection - Performance check, 250 hours; continuation tests, 1,750 hours.
Test temperature - 125°C.
Test potential - 200 percent of rated voltage.
IR - at 25°C; not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.
IR - at 125°C, not less than 1,000 megohms or 10 megohm-microfarads, whichever is less.
Cap. - Change not greater than 2 percent for voltage temperature limit T, 5 percent for limit U, and 20 percent for limit V.
DF - Not greater than initial requirement.

QUALITY ASSURANCE PROVISIONS

Sampling and inspection: MIL-C-11015 and as specified herein.

Qualification inspection: MIL-C-11015 on the respective non-TX version. (The TX version shall be procured from sources listed in QPL-11015 for the respective non-TX version.)

Quality conformance inspection: MIL-C-11015 and as follows:

Group A inspection: MIL-C-11015.
For the TX version power-stress conditioning shall be performed prior to the test specified under this group and the AQL shall be 2.5 (percent defective) minor.

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TABLE II. Voltage-Temperature Limits, Dissipation Factor

Capacitance Change with Reference to 25°C
VOLTAGE-TEMPERATURE LIMITS (%)

VOLT-TEMP SYMBOL	Steps A to D, Incl of table VIII (%)	Steps E to G, Incl of table VIII (%)	DISSIPATION FACTOR (Max %)
T	+2, -10	+2, -10	1.0
U	+2, -15	+2, -15	1.5
V	+20, -45	+20, -50	3.0

Groups B and C inspection: MIL-C-11015. Successful performance of groups B and C inspection for the respective non-TX version will suffice to demonstrate conformance to the requirement for TX version.

Initial tests: IR, capacitance, and DF shall be measured. Items that are out of capacitance tolerance, DF, and IR limits shall be removed from the lot.

Power-stress conditioning: Each capacitor intended for delivery as a TX part shall be subjected continuously to three times the rated voltage for 96 hours at an ambient temperature of 25°C.

Post power-stress conditioning measurement: After conditioning, each capacitor shall be measured for IR, capacitance, and DF. Items that fail to meet the initial requirements shall be removed from the inspection lot, and the quantity removed noted on the permanent lot history. Only lots having not more than 5 percent rejects or 1 capacitor, whichever is greater, as a result of this test shall be approved for delivery.

Acceptance: Acceptance of TX parts shall be based on the qualification, quality conformance inspection, sampling criteria of MIL-C-11015, and power-stress conditioning.

Certification: Each lot delivered as TX parts shall be accompanied by a certificate verifying that power-stress conditioning has been performed and satisfactorily met.

Traceability: Each lot intended for delivery as TX parts shall be assigned a unique lot number prior to any tests. The lot number shall identify its date of manufacture. The manufacturer shall maintain a chronological file of these lot numbers and the data records shall be made available on request of the procuring activity. The data records shall be retained by the supplier for a period of 3 years.

Qualification: MIL-C-11015, except the activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: ASNPS-20, Wright-Patterson Air Force Base, Ohio 45433. However, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, Attn: DESC-EP, 1507 Wilmington Pike, Dayton, Ohio 45420.

Marking: Non-TX parts in accordance with MIL-C-11015. TX parts shall be marked as shown in the following example:

Example:

XXXX----Mfr's symbol
CK31BT } Type designation
103KTX }

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Air Force - 11, 17, 99

(Project 5910-F299)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-11015/25B (USAF) Capacitors, Fixed, Ceramic-Dielectric

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER FSC 5910

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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the corrective action has been taken. Group C inspec-